Transpose Matrix

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The problem

- Flip a 2D matrix over its main diagonal.

Intuition

- Transposing swaps rows and columns, so the cell at row i, column j moves to row j, column i.

Approach

- 1. Allocate a matrix with the dimensions swapped
- 2. Copy each cell to its mirrored position
- 3. Return the new matrix

Java solution

```java
public class TransposeMatrix {
    public static int[][] transposeMatrix(int[][] matrix) {
        int rows = matrix.length, cols = matrix[0].length;
        int[][] out = new int[cols][rows];
        for (int r = 0; r < rows; r++) for (int c = 0; c < cols; c++) out[c][r] = matrix[r][c];
        return out;
    }
}
```

Complexity

- Time: O(w*h)
- Space: O(w*h)

Recap

- Transposing swaps rows and columns, so the cell at row i, column j moves to row j, column i.
- Complexity: O(w*h)